Managing Sustainable Tourism in Juneau

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1 Abstract

Tourism supports Juneau's economy but concentrates **1.6 million cruise visitors** a year on its roads, glacier, and neighborhoods. We build a **closed-loop equilibrium model** in which annual visitor volume is an outcome of policy, not a choice: taxes affect demand, receipts fund investment, and investment changes capacity, pressures, and resident acceptance, which feed back into demand.

Model 1 combines saturating investment responses with environmental, social, and infrastructure pressure indices, enforces 14 hard limits, and penalizes three soft targets. NSGA-II and SLSQP search all 32 combinations of five management measures, and Brent's method solves the demand equilibrium at every evaluation. The recommended policy sets an **aggregate tax lever of 14.72%** and adopts **alcohol restrictions, shore power, and traffic management**. It yields **1.472 million visitors** and **USD 345 million** in tourism activity, with **resident acceptance 0.864** and every hard limit satisfied. A **20-year test** of the fixed policy shows the peak-day limit holds for two decades if demand grows **at most 0.4% a year**, while rising real costs call for a **cost-indexed reserve**.

Model 2 traces the money. Receipts of **USD 50.78 million** pay for maintenance and measures before funding **USD 34.13 million of new projects**, of which **54.4% goes to environmental protection**. The tax lowers demand by 14.6 log points; investment recovers 6.4 through added capacity and higher acceptance.

Model 3 finds that the **revenue target, not the weights, sets the visitor scale**: without it, the preferred volume falls to 1.277 million. Revenue calibration and seasonal concentration are the most sensitive parameters. A minimax-regret search keeps the same measures and volume while **cutting worst-case regret by 6.2%**.

Model 4 transfers the framework to **Venice**. Pressure profiles reverse the lever priorities: Juneau favors shore power and capacity, while Venice favors a **daily cap, rent subsidy, and a higher fee**. Recalibrated thresholds raise the acceptance target to 0.889. Venice's 2024 access-fee data imply a **fee elasticity of about** -0.55, close to the -0.5 assumed for Juneau.

Model 5 redistributes visitors among four attractions. Investing **USD 5 million a year in less-visited sites** lowers **glacier utilization from 0.95 to 0.797** without reducing total visitors, and relieves the port–glacier road, the peak-day bottleneck. The exact margin depends on assumed investment efficiencies.

The results support structured policy comparison; behavioral responses need local calibration before implementation.

Keywords: sustainable tourism; demand equilibrium; diminishing returns; budget allocation; multi-objective optimization; TOPSIS; gravity model

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2 Introduction

2.1 Problem background

Juneau, Alaska, has approximately 30,000 residents and received about 1.6 million cruise passengers in 2023. The problem statement associates tourism with approximately USD 375 million in economic activity and describes a busiest day with seven large cruise ships and about 20,000 visitors. [1] Such concentrated arrivals place demands on roads, terminals, waste management, and public spaces. Pressure on sensitive attractions, including the Mendenhall Glacier area, adds an environmental dimension to the management problem.

Tourism benefits and tourism-related costs respond to the same policy. Higher visitor charges can finance public services but may discourage travel. Infrastructure can reduce congestion while attracting more demand. Community programs can ease local pressure and improve acceptance. A useful model must account for these feedbacks when comparing policy choices.

Tourism-related pressure	**Management response evaluated**
Concentrated arrivals and crowded public facilities	Daily visitor controls, traffic management, and infrastructure investment
Environmental impacts at ports and natural attractions	Shore power and environmental investment
Noise, congestion, and pressure on local services	Alcohol restrictions and community investment
Uneven visitor distribution and demand for public funding	Redistribution marketing and an aggregate tourism tax or fee

Local management cannot, by itself, determine glacier retreat or eliminate climate-related risks. We therefore use a normalized environmental pressure index to compare modeled tourism impacts; it is not a physical prediction of glacier change.

2.2 Problem statement

The problem calls for a management strategy that balances tourism benefits with environmental protection and residents' well-being. We organize the required work into three tasks:

1. **Build a sustainable tourism model.** Define the decision variables, objectives, and constraints, and explain how tourism-related revenue finances measures that affect the tourism system.

2. **Examine policy sensitivity.** Identify how uncertain parameters and different priorities alter the recommendation and expose constraints that limit its reliability.

3. **Extend the framework.** Adapt the model to another destination, assess ways to redirect visitors toward less-visited attractions, and communicate practical recommendations in a one-page memorandum.

2.3 Our work

Our framework links five models. M1 determines Juneau's visitor equilibrium and policy trade-offs; M2 accounts for funding and investment feedback; M3 examines parameter uncertainty and competing preferences. M4 transfers the framework to Venice. M5 uses the preference scenarios examined in M3 to study redistribution among attractions.

Figure 1 summarizes the core feedback loop. Figure 2 outlines each model's method and output. M1–M3 support Juneau's integrated policy recommendations, and M3's preference scenarios inform M5's spatial analysis.

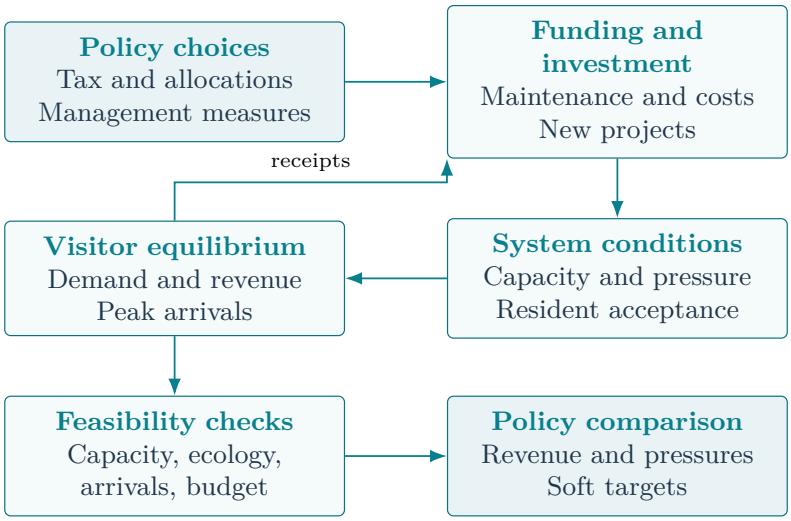


Figure 1: Core workflow. Visitor demand and policy funding are solved together.

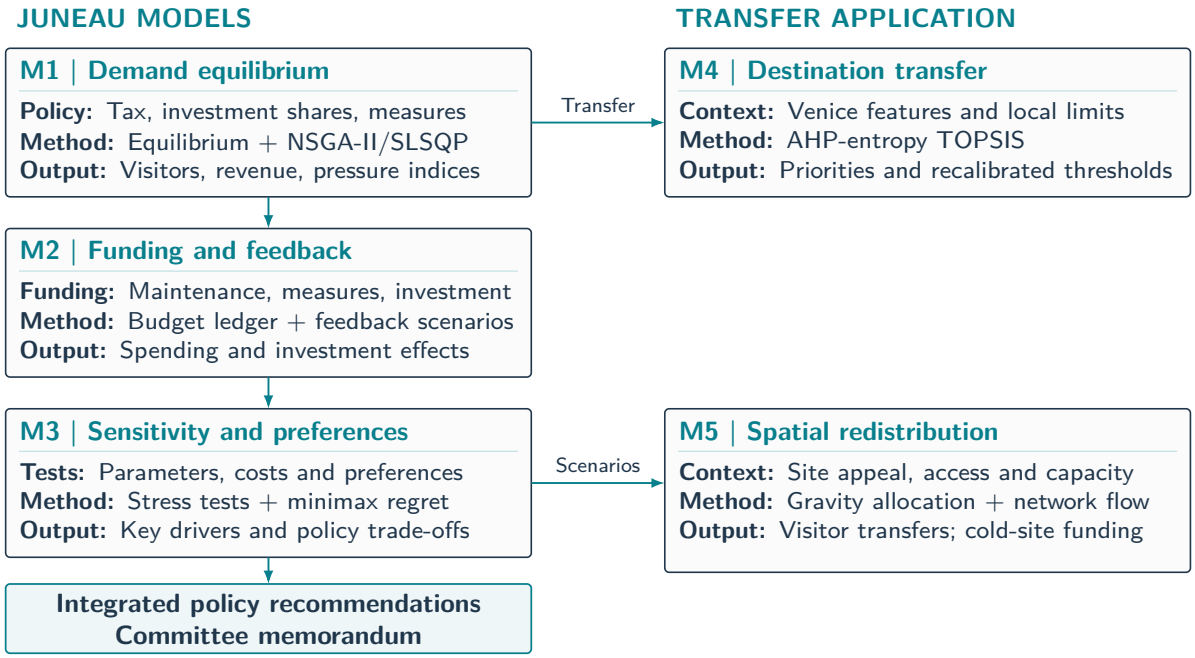


Figure 2: Model overview. M1–M3 support integrated policy recommendations.

3 Assumptions and Justifications

Assumptions	Justifications
A1. Consistent statistical baseline. The main calculation uses the problem's 2023 visitor and economic-activity figures. Supplemental reports provide context and cost proxies.	Mixing passenger counts and expenditure totals with different coverage would change spending per visitor without a defensible adjustment. We retain one numerical baseline and identify assumptions separately.
A2. Annual equilibrium and a fixed seasonal profile. Policies apply to a representative normal operating year. Annual visitors are endogenous, and the peak day is proportional to annual volume.	Annual aggregation makes the policy comparison tractable. A fixed concentration factor preserves the observed baseline peak. It does not describe individual ship schedules or unexpected disruptions.
A3. Saturating investment effects. New investment increases capacity or reduces pressure with diminishing marginal returns. Its annual-equivalent effect is available within the planning period.	Finite response ceilings represent implementation and physical limits. This is a steady operating comparison; construction delays and first-year financing require a separate implementation schedule.
A4. A single budget with separate uses. Eighty percent of modeled tax receipts may fund maintenance, adopted measures, and new investment. Each expenditure is charged once.	The allocation must be affordable at the visitor equilibrium it produces. The 80% spending cap is a planning assumption, and the maintenance reserves are proxies drawn from selected budget-proposal items.
A5. Hard limits and soft targets have distinct roles. Capacity, ecology, arrivals, tax bounds, and the budget determine feasibility. Revenue, acceptance, and social-pressure targets enter the objective as penalties.	A shortfall in a policy target should remain visible without being labeled physically impossible. The main soft targets are USD 345 million, acceptance 0.70, and social pressure 1.0.
A6. Response parameters describe scenarios. Demand elasticities, policy effects, investment parameters, and the acceptance response are declared assumptions rather than fitted relationships.	The available aggregate observations do not identify these behavioral effects. The resident index is a model output, not an estimated percentage of satisfied residents.

4 Notations

Visitor quantities use 10^4 people and monetary quantities use 10^8 USD unless stated otherwise. All pressure indices, response fractions, and scores are dimensionless. Local variables and parameters omitted from the general notation tables are defined before the relevant equations.

Sets and indices

Symbol	Meaning
$j \in \mathcal{J} = \{1, \ldots, 5\}$	Management measure index
$k \in \mathcal{K} = \{I, E, C\}$	Infrastructure, environmental, and community investment categories
$\ell \in \{R, E, S, I\}$	Revenue, environmental, social, and infrastructure score components
$h = 1, \ldots, 7;\ v = 1, \ldots, 7$	Pressure features and policy levers compared across destinations (M4)
$i \in \mathcal{I} = \{1, \ldots, 4\}$	Attraction nodes: port, glacier, whale watching, rainforest (M5)

Variables and derived quantities

Symbol	Meaning	Unit
τ	Aggregate tax or fee lever	Dimensionless
$\delta_I, \delta_E, \delta_C$	Shares of total receipts allocated to new investment	Dimensionless
m_j	Binary adoption indicator for measure j	0 or 1
$x,\ x_d$	Annual visitors and derived peak-day visitors	10^4/year; 10^4/day
$R,\ B,\ A$	Tourism economic activity, tax receipts, and pool for new investment	10^8 USD/year
$u_k,\ g_k$	New investment and its response fraction	10^8 USD/year; dimensionless
I_{cap}	Effective annual infrastructure capacity	10^4/year
$C_{env}, C_{soc}, C_{inf}$	Environmental, social, and infrastructure pressures	Dimensionless
S	Resident acceptance index	Dimensionless
$P_{\text{soft}},\ J_w$	Soft-target penalty and sustainability score	Dimensionless

Baseline quantities

Symbol	Value	Definition and status
P	3	Resident population (10^4); problem baseline
$x_0,\ x_{d0}$	160; 2	Annual and peak-day visitors; problem baseline
R_0	3.75	Tourism economic activity (10^8 USD); problem baseline
$p_0 = R_0/x_0$	0.0234375	Economic activity per 10^4 visitors (10^8 USD)
$I_0,\ S_0$	160; 0.70	Reference capacity and acceptance
τ_0	0.11	Reference aggregate tax lever
$B_{\text{pol}}, B_{\text{phys}}$	5; 7	Daily ship-policy limit and observed peak proxy

5 Models design for M1, M2, M3

5.1 Data resources and numerical conventions

We use the problem statement as the main numerical source: 1.6 million annual cruise visitors, USD 375 million in tourism economic activity, 30,000 residents, and a peak of about 20,000 visitors. Consequently, $p_0 = R_0/x_0 = 0.0234375$ in model units, or USD 234.375 per visitor. This is an aggregate economic-activity proxy, not individual taxable expenditure measured from a visitor sample.

The McKinley economic-impact report and resident survey provide supporting context. [2, 3] We do not substitute the report's approximately 1.67 million passengers or broader economic totals into the problem-based ratio. Likewise, the resident survey motivates a social dimension but does not estimate $S_0 = 0.70$ or the model's acceptance slope.

Official budget materials supply narrow cost proxies. The FY24 passenger-fee proposal lists USD 441,900 for trail, open-space, and restroom maintenance, USD 200,000 for street cleaning and repair, and USD 613,400 for emergency-service support. [4] We group the first two as infrastructure maintenance and the third as community maintenance. The resulting reserves are USD 0.6419 million and USD 0.6134 million, respectively.

5.2 Model 1: Closed-loop equilibrium model

M1 links policy choices to visitor demand and to the funding that demand generates. Annual visitor volume is not chosen directly: it is the equilibrium of a feedback loop (Section 5.2.3). Feasibility and the sustainability score are evaluated at that equilibrium.

5.2.1 Policy variables and management channels

M1 chooses a tax lever τ, three investment shares $(\delta_I, \delta_E, \delta_C)$, and five binary switches m_j. The lever τ is the fraction of tourism economic activity R collected as receipts, $B = \tau R$; it also lowers demand through M1-4. The share δ_k of B funds new investment in category k (M1-2). Setting $m_j = 1$ adopts measure j and charges its cost:

Switch	**Measure adopted when $m_j = 1$ (otherwise inactive)**
m_1	Additional daily visitor cap, stricter than the ordinary limit
m_2	Alcohol-sales restriction
m_3	Shore power for cruise ships
m_4	Traffic management
m_5	Marketing to redistribute visitors

5.2.2 Budget ledger and investment response

Tax receipts and the investable pool.

Symbol	Meaning
$R = p_0 x,\ B = \tau R$	Annual tourism economic activity and modeled tax receipts, in 10^8 USD
$\delta_{\text{total}} = 0.8$	Maximum spendable share of total receipts
$\mu_I = 0.006419$	Infrastructure maintenance reserve, in 10^8 USD
$\mu_C = 0.006134$	Community maintenance reserve, in 10^8 USD
$c_j,\ A$	Annual measure cost and residual pool for new investment, in 10^8 USD

$$R = p_0 x, \qquad B = \tau R,$$
$$A = \delta_{\text{total}} B - \mu_I - \mu_C - \sum_{j=1}^{5} c_j m_j. \tag{M1-1}$$

Visitors generate economic activity R through the spending proxy p_0, and the tax lever turns it into receipts B. Of these, $\delta_{\text{total}} B$ may be spent: maintenance $\mu_I + \mu_C$ and adopted-measure costs are deducted first, leaving the pool A for new investment. The main scenario funds this from total receipts B, not only from the increase over $B_0 = \tau_0 R_0$.

Saturating investment response.

Symbol	Meaning
$\delta_k,\ u_k$	Share of receipts and new investment in category $k \in \{I, E, C\}$, in 10^8 USD
η_k, L_k, K_k	Efficiency, ceiling parameter, and monetary saturation scale
(η_I, L_I, K_I)	$(0.7, 0.5, 0.15)$
(η_E, L_E, K_E)	$(0.8, 0.9, 0.15)$
(η_C, L_C, K_C)	$(0.6, 0.8, 0.08)$

$$u_k = \delta_k B, \qquad g_k(u_k) = \eta_k L_k \left(1 - e^{-u_k/K_k}\right), \quad k \in \{I, E, C\}. \tag{M1-2}$$

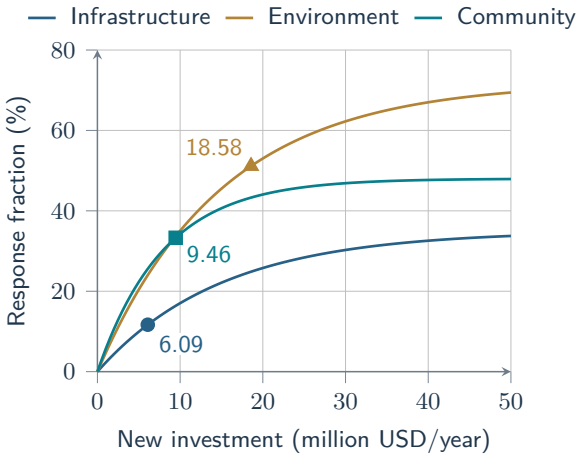


Figure 3: Investment response curves; markers show recommended spending (million USD/year).

The ratio u_k/K_k measures spending against each category's saturation scale, and $1 - e^{-u_k/K_k}$ gives diminishing returns: an additional dollar changes the response more at low spending than at high spending. The factor $\eta_k L_k$ sets the ceiling each response approaches.

Figure 3 compares the three curves on a common scale. Environmental projects have the highest ceiling (72%), community programs saturate fastest because K_C is smallest, and infrastructure has the lowest ceiling (35%).

5.2.3 Capacity, peak day, and demand equilibrium

Effective capacity and seasonal concentration.

Symbol	Meaning
$I_0 = 160,\ I_{cap}$	Reference and effective annual capacity, in 10^4 visitors/year
$T = 214,\ q = 2.675$	Assumed season length and baseline-matched peak-to-average factor
$x_d,\ x_{d0} = 2$	Modeled and baseline peak-day volume, in 10^4 visitors/day

$$I_{cap} = I_0(1 + g_I), \qquad x_d = \frac{q}{T}x = \frac{x}{80}. \tag{M1-3}$$

Infrastructure investment adds the capacity $I_0 g_I$ to the reference I_0, which matches the baseline volume. The peak day is a fixed multiple q of the seasonal average x/T; choosing q so that $qx_0/T = x_{d0}$ reproduces the observed peak and gives $x_d = x/80$.

Demand response to policy.

Symbol	Meaning
$x_0 = 160,\ \tau_0 = 0.11$	Reference annual visitors and aggregate tax lever
$\varepsilon = -0.5,\ \alpha = \beta = 0.2$	Tax, capacity, and acceptance elasticities
$S,\ S_0 = 0.70$	Current and reference resident acceptance indices
$(\kappa_1, \ldots, \kappa_5)$	Direct demand effects $(-0.10, -0.05, 0.02, 0.03, 0.10)$

$$x = x_0 \left(\frac{\tau}{\tau_0}\right)^{\varepsilon} \left(\frac{I_{cap}}{I_0}\right)^{\alpha} \left(\frac{S}{S_0}\right)^{\beta} \prod_{j=1}^{5}(1 + \kappa_j m_j). \tag{M1-4}$$

Every factor equals 1 in the reference state, so demand returns to x_0 at $\tau = \tau_0$ with no investment, no measures, and $S = S_0$. A higher tax lowers demand; added capacity and higher acceptance raise it; κ_j are the measures' direct effects. Section 5.3 separates these contributions at the recommendation.

Solving for the equilibrium volume. Visitor volume appears on both sides of M1-4. A trial volume x sets receipts $B = \tau p_0 x$, which fund g_I and g_C; these change I_{cap}, C_{soc}, and hence S, and so the demand that results. Writing the right-hand side as $\Phi(x)$, the equilibrium is the fixed point

$$x^* = \Phi(x^*) \quad \Longleftrightarrow \quad f(x^*) = x^* - \Phi(x^*) = 0. \tag{M1-4a}$$

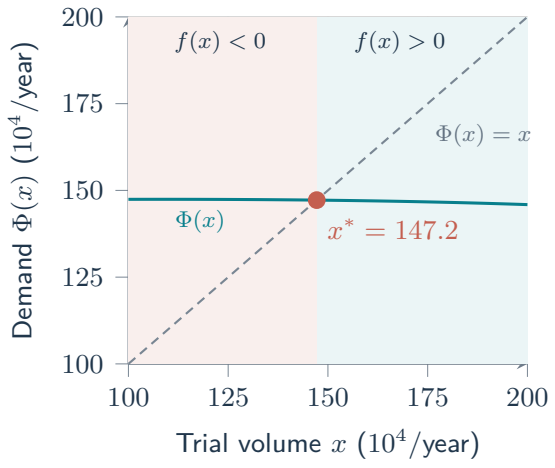


Figure 4: Demand $\Phi(x)$ for the recommended policy and its fixed point.

Brent's method [7] finds the root of f on the bracket $[0, 10^4]$. It combines bisection with secant and inverse-quadratic steps, so it keeps the guarantee of bisection while converging faster.

Figure 4 shows the recommended policy. Φ is nearly flat, with slope about -0.01 near x^*: added receipts raise capacity while added crowding lowers acceptance, and both enter with small elasticities ($\alpha = \beta = 0.2$). The curve therefore crosses the line $\Phi(x) = x$ exactly once, at $x^* = 147.2$, where the residual is below 10^{-13}.

5.2.4 Environmental, social, and infrastructure pressures

The indices equal 1 when evaluated at $x = x_0$, $x_d = x_{d0}$, $I_{cap} = I_0$, with no measures and no new investment. This is a common accounting reference; the reference state need not itself fund the modeled maintenance reserves.

Environmental pressure.

Symbol	Meaning
$e_1 = e_2 = 0.5$	Weights on annual activity and peak-day pressure
$\kappa_{\text{pow}} = 0.30$	Reduction in the peak component under shore power
g_E	Environmental investment response from M1-2

$$C_{env} = \left[e_1 \frac{x}{x_0} + e_2 \frac{x_d}{x_{d0}} (1 - \kappa_{\text{pow}} m_3) \right] (1 - g_E). \tag{M1-5}$$

Environmental burden combines annual activity and peak-day arrivals, each relative to its baseline. Shore power reduces only the peak component, which reflects ships at berth; environmental investment then scales down the total.

Social pressure.

Symbol	Meaning
$(s_h, s_n, s_{nz}) = (0.5, 0.3, 0.2)$	Weights on crowding, peak congestion, and noise
$\kappa_{\text{tr}} = 0.40, \kappa_{\text{alc}} = 0.20$	Assumed traffic and alcohol-restriction effects
g_C	Community investment response from M1-2

$$\begin{aligned} C_{soc} = \Bigg[& s_h \left(\frac{x}{x_0} \right)^2 + s_n \frac{x_d}{x_{d0}} (1 - \kappa_{\text{tr}} m_4) \\ & + s_{nz} (1 - \kappa_{\text{alc}} m_2) \Bigg] (1 - g_C). \end{aligned} \tag{M1-6}$$

Crowding grows with the square of annual volume, so each extra visitor adds more burden than the last. Traffic management reduces peak congestion, alcohol restrictions reduce noise, and community investment scales down the combined burden.

Infrastructure pressure.

Symbol	Meaning
$i_p = 0.6,\ i_d = 0.4$	Weights on annual capacity utilization and peak-day arrivals
I_{cap}	Effective capacity after infrastructure investment

$$C_{inf} = i_p \left(\frac{x}{I_{cap}} \right)^2 + i_d \frac{x_d}{x_{d0}}. \tag{M1-7}$$

Annual pressure rises with the square of utilization x/I_{cap}, so infrastructure investment acts through I_{cap}. The second term is peak-day load on terminals relative to the baseline peak.

5.2.5 Resident acceptance and demand feedback

The boundary effect.

Symbol	Meaning
$S_0 = 0.70,\ \gamma = 2$	Reference acceptance and response strength
C_{soc}	Social pressure from M1-6
$\sigma(z) = (1 + e^{-z})^{-1}$	Sigmoid response function

$$S = \sigma \left(\ln \frac{S_0}{1 - S_0} - \gamma (C_{soc} - 1) \right). \tag{M1-8}$$

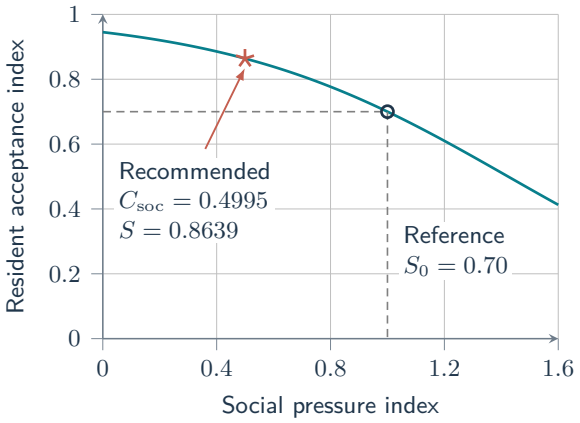


Figure 5: Logistic relationship between social pressure and resident acceptance.

The term $-\gamma(C_{soc} - 1)$ makes lower social pressure raise acceptance, and $\ln[S_0/(1-S_0)]$ places the reference $S = S_0$ at $C_{soc} = 1$.

The sigmoid creates a boundary effect: its slope $\gamma S(1 - S)$ shrinks as S nears 0 or 1. The same reduction in crowding, congestion, or noise therefore yields a larger gain at moderate acceptance than near an extreme. At the recommendation (Figure 5) the slope is 0.24, against 0.42 at the reference. Acceptance then feeds back into demand through $(S/S_0)^{\beta}$ in M1-4.

5.2.6 Hard constraints and soft-target penalties

The implementation checks 14 hard inequality residuals, grouped below into six physical and operating checks, two fiscal checks, and six domain checks. The demand equation is a separate equality. Binary adoption is enforced through the policy representation and exhaustive enumeration during refinement.

Physical and operating limits: six checks.

Symbol	Meaning
$E_{\max} = 1.0$	Environmental-pressure ceiling
$\Delta_{cap} = 0.4$	Extra daily-cap reduction, in 10^4 visitors/day
$\bar{c} = 0.4,\ B_{\text{pol}} = 5$	Assumed passengers per ship (10^4) and daily ship-policy limit
$B_{\text{phys}} = 7,\ T = 214$	Observed ship-peak proxy and assumed season length

$$\begin{aligned} x &\le I_{cap}, & C_{env} &\le E_{\max}, \\ x_d &\le x_{d0} - \Delta_{cap} m_1, & x_d &\le \bar{c} B_{\text{pol}}, \\ B_{\text{pol}} &\le B_{\text{phys}}, & x_d &\ge x/T. \end{aligned} \tag{M1-9}$$

Annual visitors must fit effective capacity, and environmental pressure must stay below its ceiling. The peak day must respect the ordinary limit, tightened by Δ_{cap} when $m_1 = 1$, and the ship-policy ceiling $\bar{c}B_{\text{pol}}$, which in turn may not exceed the observed ship peak. The last condition keeps the peak day at or above the seasonal average.

Fiscal limits: two checks.

Symbol	Meaning
$A,\ B$	Investable pool and total modeled receipts from M1-1
$\delta_k,\ u_k = \delta_k B$	Receipt shares and derived new investment amounts

$$\sum_{k \in \mathcal{K}} u_k \le A, \qquad \sum_{k \in \mathcal{K}} \delta_k \le \delta_{\text{total}}. \tag{M1-10}$$

New investment may not exceed the pool left after maintenance and measures, and the investment shares may not exceed the spendable share of receipts.

Domain limits: six checks. The remaining checks bound the variables to their admissible ranges:

$$0 \le S \le 1, \qquad \tau_{\min} = 0.09 \le \tau \le \tau_{\max} = 0.15, \qquad 0 \le \delta_k \le 1 \ \ (k \in \mathcal{K}). \tag{M1-11}$$

Three soft targets.

Symbol	Meaning
$S_{\min} = 0.70,\ R_{\min} = 3.45$	Acceptance target and revenue target (10^8 USD)
$H_{\max} = 1.0$	Social-pressure target ceiling
$(\lambda_S, \lambda_R, \lambda_H) = (5, 5, 2)$	Penalty coefficients

$$P_{\text{soft}} = \lambda_S \max\{0, S_{\min} - S\} + \lambda_R \max\left\{0, \frac{R_{\min} - R}{R_0}\right\} + \lambda_H \max\{0, C_{soc} - H_{\max}\}. \tag{M1-12}$$

Each term is zero once its target is met and grows linearly with the shortfall. Dividing the revenue gap by R_0 puts it on the same dimensionless scale as the score.

5.2.7 Sustainability score and preference weights

Symbol	Meaning
J_w	Dimensionless sustainability score
w_R, w_E, w_S, w_I	Nonnegative preference weights summing to 1
$R/R_0,\ C_{env}, C_{soc}, C_{inf}$	Normalized revenue benefit and the three pressure costs

$$\max J_w = w_R \frac{R}{R_0} - w_E C_{env} - w_S C_{soc} - w_I C_{inf} - P_{\text{soft}}. \tag{M1-13}$$

The score rewards normalized economic activity, subtracts the three weighted pressures, and subtracts the penalties for unmet targets.

The preference vector is

$$w = (w_R, w_E, w_S, w_I) = (0.3165168,\ 0.2750766,\ 0.2042066,\ 0.2042000).$$

Revenue receives the highest priority, environmental protection the next, and social and infrastructure pressures almost equal priority. For example, a 0.01 increase in R/R_0 raises the score by about 0.00317, while a 0.01 reduction in C_{env} raises it by about 0.00275.

5.2.8 Solution procedure

The search varies τ, the investment shares, and m while holding w fixed. Figure 6 outlines the procedure: NSGA-II [5] explores trade-offs, SLSQP [6] refines all 32 measure combinations, and Brent's method [7] solves each visitor equilibrium. Feasible policies are ranked by J_w, followed by independent equilibrium, budget, and constraint checks.

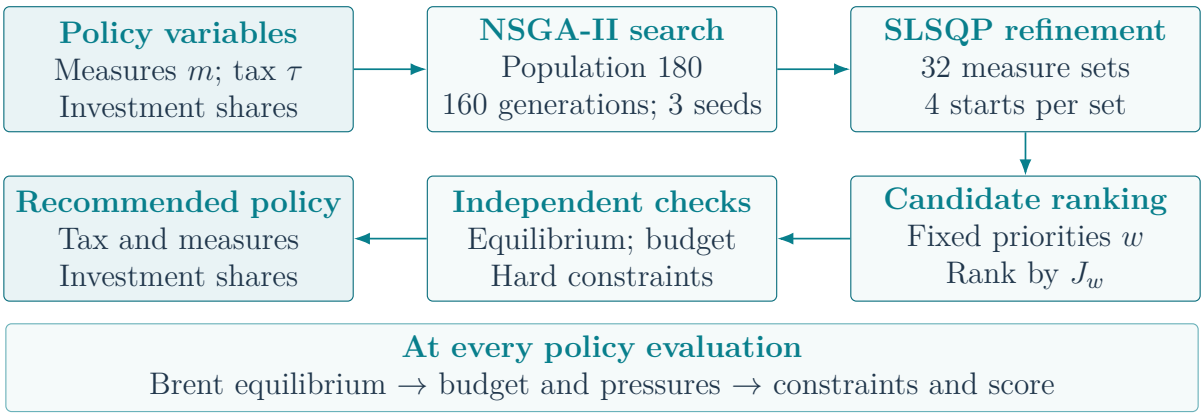


Figure 6: Search, refinement, and verification of the M1 policy.

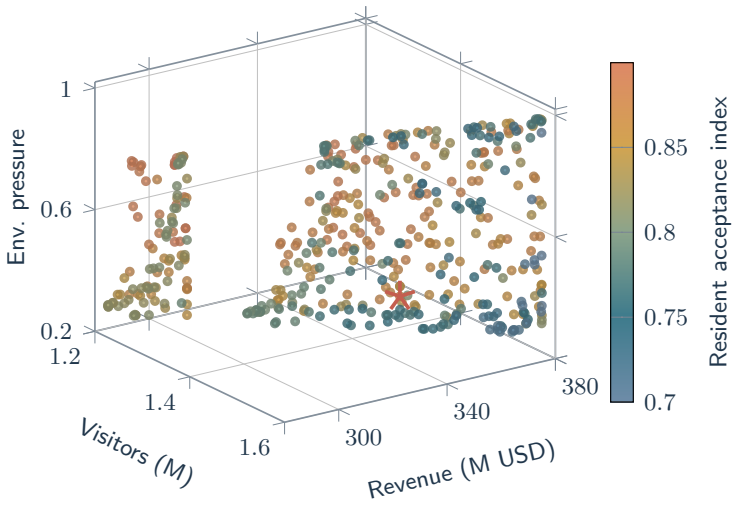


Figure 7: Approximate Pareto set; color shows resident acceptance and the star marks the recommendation.

Figure 7 shows the 458 saved near-nondominated candidates. It makes the range of feasible trade-offs visible before the fixed-weight recommendation is reported.

The recommended policy (1.472 million visitors, USD 345 million) lies in the low-pressure part of the set. Visitor volume and revenue are linked by $R = p_0 x$, so the plot is a projection of the four-objective candidate set.

5.2.9 Recommended policy and budget

Under the main assumptions and preference vector, the best reported candidate adopts alcohol restrictions, shore power, and traffic management, $m = (0, 1, 1, 1, 0)$. Its annual volume of 1.472 million coincides with the USD 345 million revenue target, identifying the recommendation as a compromise under the stated income preference. The equilibrium outcomes are listed below.

Quantity	**Symbol**	**Value**
Annual visitors (million/year)	x	1.4720
Peak-day visitors (people/day)	x_d	18,400
Tourism economic activity (million USD/year)	R	345.000
Aggregate tax lever	τ	14.7188%
Response fractions	(g_I, g_E, g_C)	$(0.1168, 0.5113, 0.3330)$
Effective annual capacity (million/year)	I_{cap}	1.7868
Environmental pressure	C_{env}	0.3822
Social pressure	C_{soc}	0.4995
Infrastructure pressure	C_{inf}	0.7752
Resident acceptance index	S	0.8639
Sustainability score	J_w	-0.074224

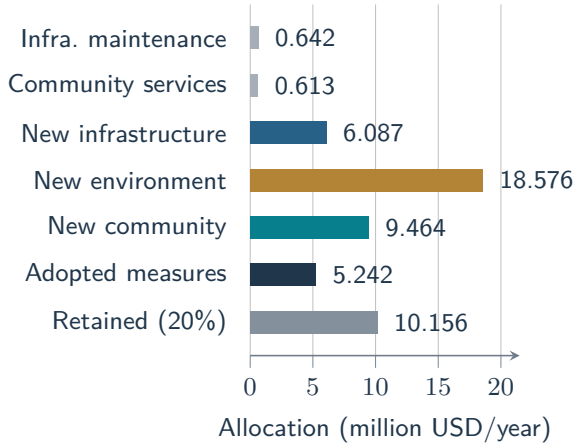


Figure 8: Allocation of annual tax receipts (million USD/year).

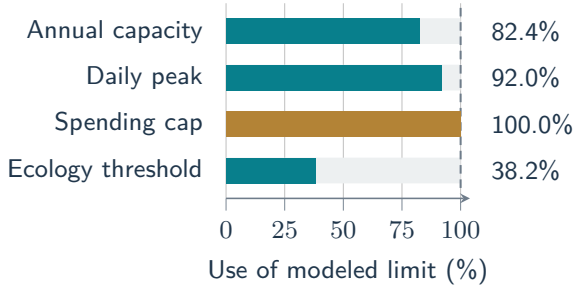


Figure 9: Constraint usage at the recommended equilibrium; the dashed line marks each limit.

Budget. Modeled tax receipts are $B =$ USD 50.78 million/year. The retained 20% (10.16) is held in reserve, and the spendable 80% (40.62) is used in full: 1.26 for maintenance, 5.24 for the adopted measures, and 34.13 for new projects (Figure 8). Because returns to each channel diminish, the optimum spreads new investment: environmental projects receive about 54.4%, with the remaining 45.6% shared by community programs and infrastructure.

Constraints. Figure 9 compares equilibrium use with each hard limit. Only the spending cap binds. Peak-day arrivals reach 92.0% of the daily limit and annual visitors 82.4% of effective capacity, while environmental pressure uses 38.2% of the ecological threshold. Further improvement is therefore limited by available funds, while capacity and ecological limits retain positive margins.

5.2.10 Twenty-year operating test

M1 is an annual steady state. To test whether the recommendation stays fundable and feasible over time, we run the budget ledger of M1-1 forward for 20 years, holding (τ, δ_k, m) fixed and starting from the recommended equilibrium. Whatever part of the investable pool A_t is not spent carries over as a reserve Q_t. Investment effects now persist and decay instead of appearing at once, and acceptance and environmental load adjust gradually:

Symbol	Meaning
$Q_t,\ Q_1 = 0$	Cumulative reserve; $Q_t < 0$ is an unfunded shortfall, not borrowing (10^8 USD)
$h_{k,t},\ \psi_k$	Surviving investment effect (replaces g_k); decay rates $(\psi_I, \psi_E, \psi_C) = (0.08, 0.20, 0.25)$
$E_t;\ \zeta = 0.35,\ \varrho = 0.15$	Environmental load; adjustment rates of acceptance and load
$\nu_x,\ \nu_c,\ \nu_E,\ e_{cl}$	Scenario growth of demand, real costs, and ecological tightening; annual climate load

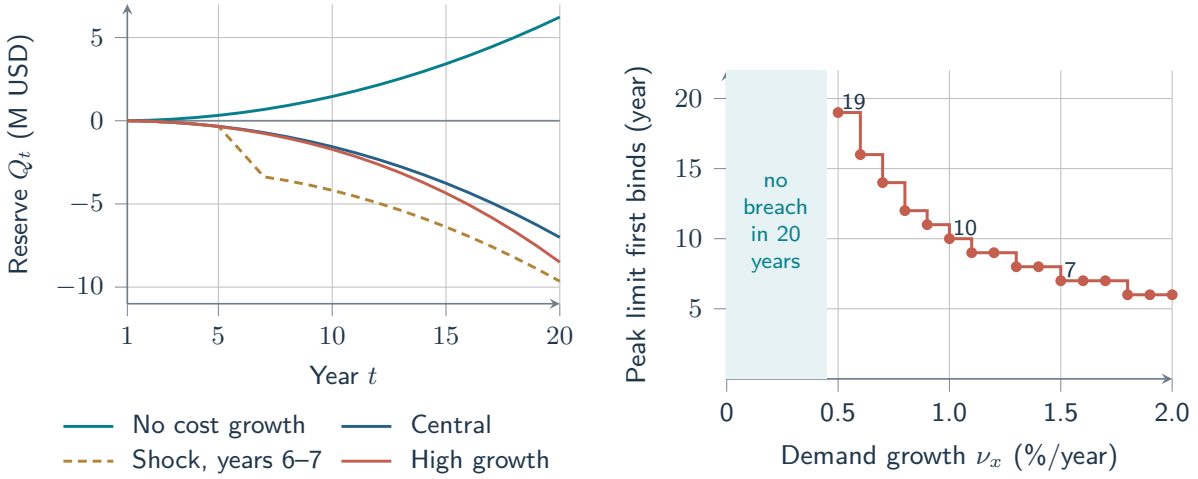
$$
\begin{aligned}
Q_{t+1} &= Q_t + A_t - \textstyle\sum_k u_{k,t}, \qquad A_t = \delta_{\text{total}} B_t - (\mu_I + \mu_C + \textstyle\sum_j c_j m_j)(1+\nu_c)^{t-1},\\
h_{k,t+1} &= (1-\psi_k) h_{k,t} + \psi_k\, g_k\big(u_{k,t}/(1+\nu_c)^{t-1}\big),\\
S_{t+1} &= (1-\zeta) S_t + \zeta\, S(C_{soc,t}), \qquad E_{t+1} = (1-\varrho) E_t + \varrho\, C_{env,t} + e_{cl}.
\end{aligned}
\tag{M1-14}
$$

Demand follows M1-4, scaled by $(1+\nu_x)^{t-1}$, with $I_{cap,t} = I_0(1+h_{I,t})$ and S_t; $S(\cdot)$ is M1-8, and $E_{\max}$ tightens by ν_E per year. With no growth, the recursion reproduces the M1 equilibrium

in every year, which checks the extension. The new rates are uncalibrated assumptions, and the scenarios are tests, not forecasts.

Scenario (ν_x, ν_c)	**Cash shortfall**	**Peak-day limit**	Q_{20} (M USD)	S_{20}
Stationary (0, 0)	none	none	0.0	0.864
Central (0.5%, 1%)	year 2	year 19	−7.0	0.848
Central with 20% demand drop in years 6–7	year 2	year 19	−9.7	0.848
High growth (1.5%, 2%)	year 2	year 7	−8.5	0.808
Central with investment decay doubled	year 2	year 19	−7.0	0.848

Table 2: First breach year and year-20 state with the M1 policy held fixed.



(a) Cumulative reserve Q_t

(b) Peak-day limit vs. demand growth (central costs)

Figure 10: Twenty-year test of the fixed M1 policy: real cost growth creates the deficit (a), and demand growth determines when the peak-day limit binds (b).

Fiscal margin. The recommendation spends the whole spending cap (Figure 9), so it leaves no slack. With real costs rising 1% a year, maintenance and measure costs outgrow the fixed shares. The shortfall starts at USD 0.03 million in year 2 and reaches USD 0.74 million a year by year 20, 1.3% of receipts. Without cost growth, the same policy instead builds a USD 6.2 million reserve (Figure 10a). A demand drop is harder: receipts and investment fall together, but maintenance and measure costs do not, so the deficit jumps to USD 1.5 million a year in the shock years.

Physical margin. Demand growth, not cost, determines when the peak-day and ship limits bind. With $x_d = x/80$, the 20,000-visitor peak corresponds to 1.6 million visitors, leaving the recommendation 8.7% of headroom. That headroom lasts the full 20 years if demand grows at most 0.4% a year, but is used up by year 19 at 0.5%, year 10 at 1%, and year 7 at 1.5% (Figure 10b). Acceptance stays above 0.80 and environmental load below 62% of the tightening ceiling in every scenario. Doubling decay barely matters, because the fixed shares keep refunding services.

The policy therefore needs two adaptive rules rather than a different design. Index investment shares to costs and hold a reserve of about USD 3 million for demand shocks. Re-solve M1 every four years, adopting the daily cap m_1 once observed growth exceeds about 0.4% a year.

5.3 Model 2: Budget allocation and investment feedback

M2 traces how tourism receipts fund public improvements and how those improvements feed back into the tourism system. It uses the policy and response functions established in M1.

5.3.1 From tax receipts to public improvements

Symbol	Meaning
U	Receipts retained after all modeled expenditure
$\mu_I, \mu_C,\ c_j m_j,\ u_k$	Maintenance reserves, adopted-measure costs, and new investment, as in M1

$$B = U + \mu_I + \mu_C + \sum_{j=1}^{5} c_j m_j + u_I + u_E + u_C. \tag{M2-1}$$

U records the retained portion of receipts. $\mu_I + \mu_C$ funds basic services before new improvements are made. $\sum c_j m_j$ pays for the measures selected by m. $u_I + u_E + u_C$ funds new infrastructure, environmental, and community projects. This identity assigns every dollar of B to one use and keeps maintenance and measure costs separate from investments that generate g_k.

For the M1 recommendation, Figure 8 already shows this identity: USD 50.78 million in receipts splits into 10.16 retained, 1.26 for maintenance, 5.24 for measures, and 34.13 for new projects. Environmental projects receive 54.4% of new investment because they reduce the environmental-pressure cost, while infrastructure and community investment support capacity and residents' living conditions.

Symbol	Meaning
$C_{env}^{\text{raw}}, C_{soc}^{\text{raw}}$	Pressures at the same visitor volume and measure settings, before new-investment reductions
g_I, g_E, g_C	Investment-response fractions defined in M1-2

$$\begin{aligned} I_{cap} &= I_0(1+g_I), \\ C_{env} &= C_{env}^{\text{raw}}(1-g_E), \qquad C_{soc} = C_{soc}^{\text{raw}}(1-g_C). \end{aligned} \tag{M2-2}$$

$I_0 g_I$ is the additional annual service capacity supplied by infrastructure investment. $C_{env}^{\text{raw}} g_E$ is the environmental-pressure reduction, leaving the fraction $1 - g_E$. Similarly, $C_{soc}^{\text{raw}} g_C$ removes part of the crowding, congestion, and noise burden, leaving $1 - g_C$. Lower social pressure then increases resident acceptance through M1-8.

Within the recommended equilibrium, capacity rises by 11.68%, while environmental and social pressures fall by 51.13% and 33.30%, respectively. Each comparison holds visitor volume and management measures at that equilibrium.

5.3.2 How investment feeds back into demand

Taking logarithms of the M1 demand equilibrium makes the contribution of each channel explicit. All symbols retain their M1 definitions.

$$\ln \frac{x}{x_0} = \epsilon \ln \frac{\tau}{\tau_0} + \alpha \ln(1+g_I) + \beta \ln \frac{S}{S_0} + \sum_{j=1}^{5} \ln(1+\kappa_j m_j). \tag{M2-3}$$

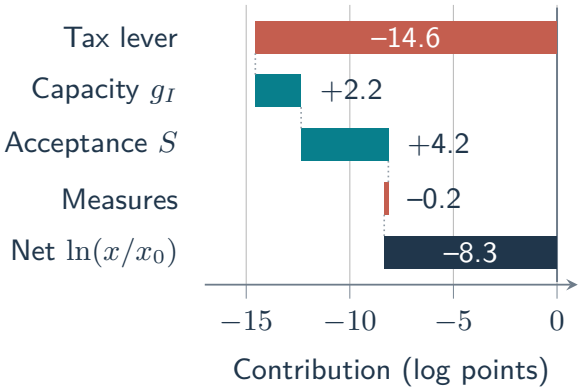


Figure 11: Channel contributions to $\ln(x/x_0)$ at the M1 recommendation (×100).

Each term is one channel's approximate percentage effect on demand. Figure 11 evaluates them at the recommendation. Raising τ from 11% to 14.72% lowers demand by 14.6 log points. Investment recovers 2.2 through capacity and 4.2 through acceptance, and the measures nearly cancel. The net −8.3 gives $x = 0.92\,x_0 = 1.472$ million.

Infrastructure and community investment therefore attract extra demand, partly offsetting their pressure relief. Changed arrivals also change $B = \tau p_0 x$, completing the funding loop. Environmental investment has no demand term and acts only on C_{env} and the score.

Restricting new-project funding to the increase in receipts, $\max(B - B_0, 0)$, leaves USD 8.693 million for new projects after measure costs, compared with USD 34.127 million under total-receipt funding. Here $B_0 = \tau_0 R_0$ continues to fund existing services. Reallocating toward a stricter acceptance target of 0.90 raises community investment to USD 19.656 million and acceptance to 0.8815; the remaining shortfall shows the trade-off between resident benefits and other spending needs.

5.4 Model 3: Sensitivity and preference robustness

M3 examines how the recommendation changes when system parameters or decision priorities change. Parameter tests assess the operating conditions faced by a policy; preference tests vary w to assess how different priorities change the preferred policy.

5.4.1 Parameter sensitivity and joint stress

Symbol	Meaning
$p = (\tau, \delta_I, \delta_E, \delta_C, m)$	Full policy vector; p^0 denotes the M1 recommendation
$\theta^0, \theta_k^{\pm}$	Baseline parameters and parameter k multiplied by 0.9 or 1.1
$p_k^{\pm}$, D_k	Readjusted policy under the perturbation and its maximum absolute score change

$$\Delta J_k^{\pm} = J_w(p_k^{\pm}; \theta_k^{\pm}) - J_w(p^0; \theta^0),$$
$$D_k = \max\{|\Delta J_k^{-}|, |\Delta J_k^{+}|\}. \tag{M3-1}$$

$J_w(p_k^{\pm}; \theta_k^{\pm})$ measures policy performance after adjusting decisions to the changed conditions. Subtracting $J_w(p^0; \theta^0)$ gives the change from the main result under the same preference vector. D_k takes the larger absolute response, identifying parameters whose uncertainty changes the attainable score most. Keeping p^0 unchanged provides a separate check of whether the original policy remains feasible.

Among the 12 parameters tested, the revenue calibration R_0 has the largest D_k, followed by seasonal concentration q. The first changes revenue per visitor, receipts, and the income-target shortfall; the second concentrates annual arrivals into busier peak days. Figure 12 shows the five largest responses. Investment efficiencies and saturation scales have smaller score effects over the tested changes.

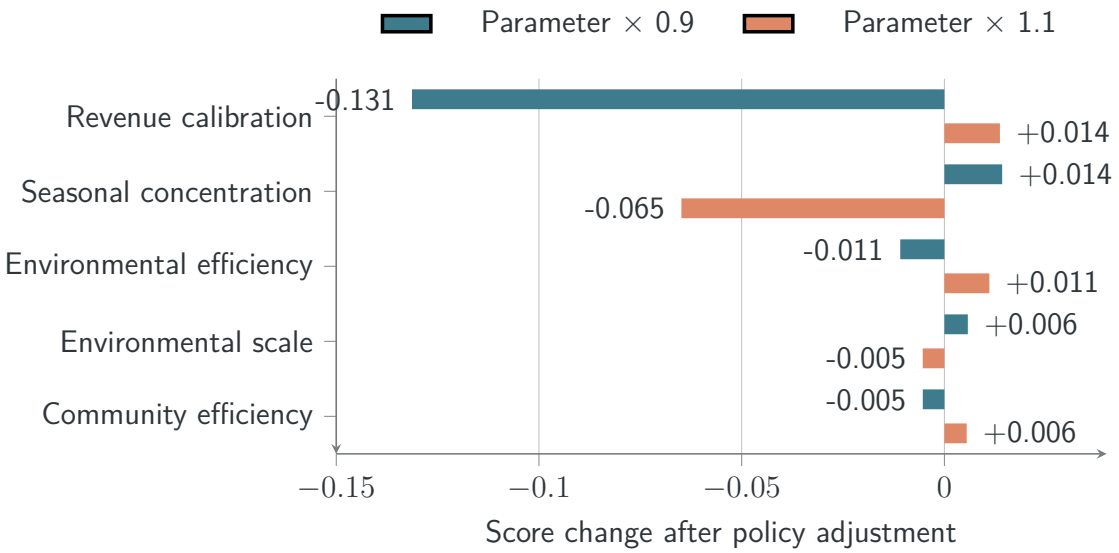


Figure 12: Largest score responses to 10% parameter decreases and increases.

Joint variation of demand elasticity, investment responses, seasonal concentration, and measure costs leaves the unchanged M1 policy feasible in 245 of 600 cases. Budget violations occur in 309 cases; the daily and ship-arrival ceilings are exceeded in the same 75 cases. These overlapping counts describe the tested conditions, rather than real-world event probabilities. They point to fiscal flexibility and peak-day management as the main needs for adaptation.

5.4.2 Revenue-target mechanism behind the sensitivity result

M2 links tourism revenue to tax receipts and investment. Section 5.4.1 then identifies the revenue calibration R_0 as the most sensitive parameter. To interpret this result, we examine how R_0 also changes the visitor volume needed to meet the revenue target. Denote this volume by x_R. From $R = p_0 x$,

$$x_R = \frac{R_{\min}}{p_0} = x_0 \frac{R_{\min}}{R_0} = 160 \times \frac{3.45}{3.75} = 147.2 \quad (10^4 \text{ visitors/year}). \tag{M3-2}$$

$R_{\min}/p_0$ converts the economic-activity target into a visitor volume. Substituting $p_0 = R_0/x_0$ shows that a lower R_0 requires more visitors to reach the same target. Below x_R, the income-shortfall penalty increases as visitor volume falls; once the target is met, this penalty becomes zero. The target therefore changes the marginal benefit of additional visitors.

For the 10% decrease in R_0 tested above, meeting the unchanged revenue target would require 1.636 million visitors, exceeding the 1.600 million annual volume allowed by the peak-arrival ceiling. The resulting income shortfall adds a penalty even after the policy is readjusted; lower receipts also reduce investment funding. These two channels explain the strong adverse response to this perturbation.

Under the baseline calibration, removing only the income-shortfall penalty lowers the recommended volume from 1.472 to approximately 1.277 million visitors. This comparison confirms that the revenue target actively influences the recommended scale.

5.4.3 Preference changes and minimax regret

In $w = (w_R, w_E, w_S, w_I)$, the components assign importance to revenue, environmental protection, residents' living conditions, and infrastructure performance.

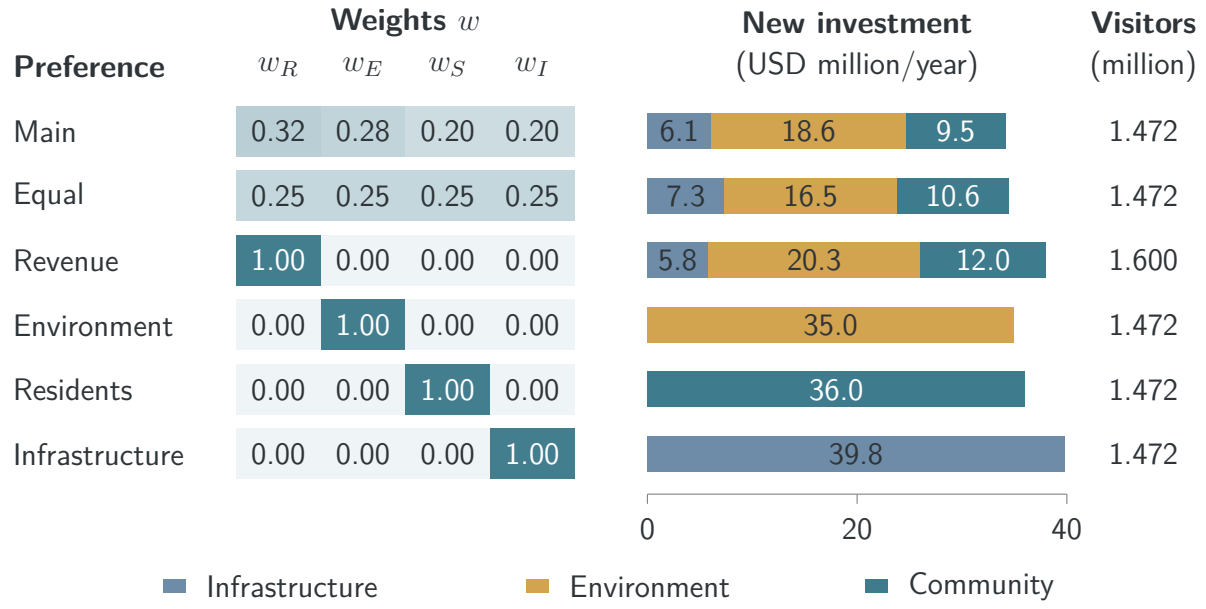


Figure 13: Representative weights and the resulting investment allocations and visitor volumes.

Across all 18 tested vectors, 13 retain approximately 1.472 million visitors; investment and measure choices can still differ. Regret compares the resulting policies under each same preference.

Symbol	Meaning
$\mathcal{W}, \mathcal{P}$	The 18 tested preference vectors and the feasible policy domain
$p^\star(w)$	Best-known policy found under preference w
$r_w(p),\ L(p),\ p_{\text{rob}}$	Regret under w, worst regret, and the minimax-regret policy

$$r_w(p) = J_w\big(p^\star(w)\big) - J_w(p), \qquad L(p) = \max_{w \in \mathcal{W}} r_w(p), \qquad p_{\text{rob}} = \arg\min_{p \in \mathcal{P}} L(p). \tag{M3-3}$$

$J_w(p^\star(w)) - J_w(p)$ measures the performance forgone by choosing p under the same weights. $L(p) = \max_w r_w(p)$ takes its largest loss across preferences. Minimizing $L(p)$ selects the policy that limits this worst loss when the committee's priorities remain unsettled.

Figure 14 shows a 6.20% reduction in maximum regret, from 0.11810 to 0.11078. The robust alternative keeps $m = (0, 1, 1, 1, 0)$ and 1.472 million visitors, with a small investment shift toward infrastructure.

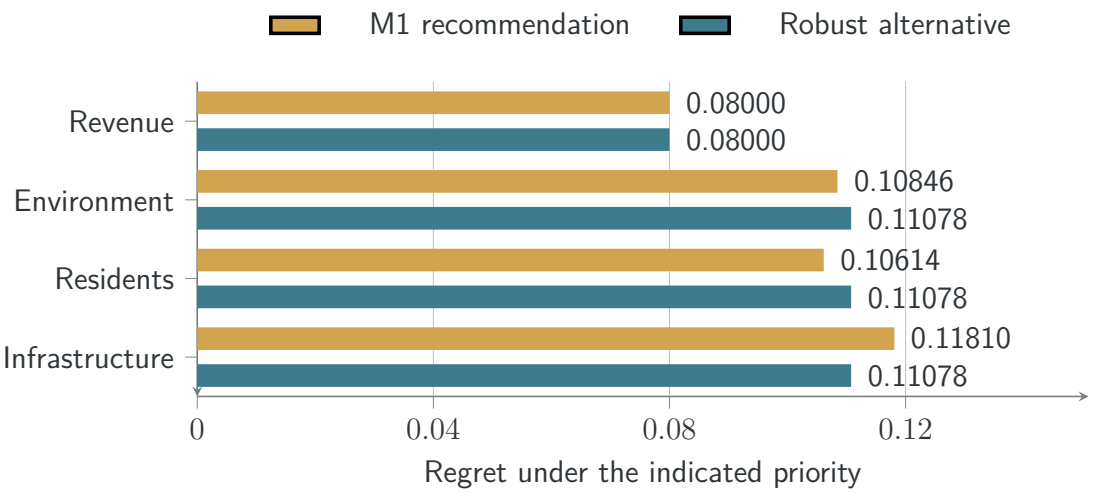


Figure 14: Regret of the main and robust policies under single-priority weights.

Preference robustness leaves the parameter-stress pass count unchanged at 245 of 600 cases.

5.5 Disadvantages in the Model

The modeling framework can be improved in the following areas as more detailed data and implementation evidence become available.

1. **Calibration of behavioral and investment responses.** Visitor records, project evaluations, and repeated resident surveys can improve estimates of demand elasticities, investment responses, and resident acceptance, and quantify uncertainty in these relationships.

2. **Seasonal and spatial resolution.** The annual model uses a fixed relationship between total visitors and peak-day arrivals. Linking it to monthly cruise schedules and attraction-level flows can capture changing arrival patterns, local bottlenecks, and the effects of targeted visitor management.

3. **Investment timing and fiscal flexibility.** The steady operating comparison can be extended to include construction delays, gradual project benefits, and first-year financing requirements. A contingency reserve and adaptive spending rules can also provide flexibility when implementation costs differ from the planning assumptions.

4. **Policy targets and solution assessment.** Employment data and public consultation can strengthen the basis for the revenue target and preference weights. Broader searches and additional optimization methods can improve coverage of nonlinear policy alternatives and the assessment of solution quality.

6 Model 4: Transferring the framework to Venice

Venice faces a different form of overtourism [8] from Juneau: dense urban crowding, housing displacement, and resident conflict in a historic centre, rather than pressure on a glacier and wilderness. M4 keeps the M1 framework unchanged. The decision vector (τ, δ_k, m_j), the pressure indices $C_{env}, C_{soc}, C_{inf}$, the acceptance index S, and the hard limits and soft targets all carry over. Venice demand and investment responses are not observed, so the equilibrium itself cannot be recalibrated. M4 therefore transfers two things semi-quantitatively: which M1 levers deserve priority, and how M1's thresholds should move. Real access-fee data then check the tax channel.

6.1 Pressure features linked to the M1 indices

Each destination $d \in \{J, V\}$ is scored on seven pressure features $F_h^d \in \{1, \dots, 5\}$ (1 negligible, 5 severe), with intensity $\tilde{F}_h^d = F_h^d/5$. Each feature loads on an M1 quantity, and each Venice score is tied to a documented fact (Table 3).

Feature	**M1 link**	F^J	F^V	**Venice evidence**	ω_h
F1 Natural environment	C_{env}	5	2	Urban historic centre; no wilderness	0.187
F2 Density and crowding	C_{soc}	2	5	3.82M historic-centre arrivals	0.164
F3 Housing displacement	S	2	5	49,693 tourist beds vs. 49,304 residents	0.206
F4 Visitor concentration	x_d	4	5	5.66M arrivals concentrated in centre	0.104
F5 Transport bottleneck	C_{inf}	3	5	Large-ship curbs; 25-person group cap	0.104
F6 Ecosystem fragility	$E_{\max}$	5	4	UNESCO-protected lagoon	0.124
F7 Resident conflict	S	3	5	Access fee justified by beds exceeding residents	0.111

Table 3: Pressure features, their M1 links, scores, Venice evidence, and combined weights.

$$\omega_h = \tfrac{1}{2}\,\omega_h^{\mathrm{AHP}} + \tfrac{1}{2}\,\omega_h^{\mathrm{ent}}, \qquad \omega_h^{\mathrm{ent}} \propto 1 + \frac{1}{\ln 2}\sum_d p_h^d \ln p_h^d, \quad p_h^d = \frac{\tilde{F}_h^d}{\tilde{F}_h^J + \tilde{F}_h^V}. \tag{M4-1}$$

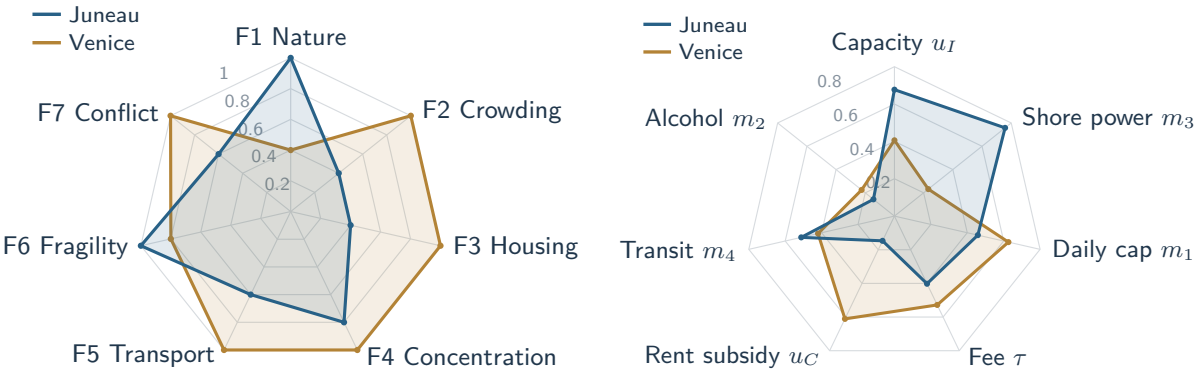
The AHP weights ω_h^{AHP} come from a 7×7 pairwise matrix [9] and give structural importance. The matrix is consistent: $\lambda_{\max} = 7.106$ and $\mathrm{CR} = 0.0134 < 0.1$. The entropy weights [10] ω_h^{ent} favor features on which the destinations differ: F1–F3 receive 0.263 each, while F4 and F6, which are severe in both, receive 0.017. With only two destinations the entropy weights are coarse, so they are averaged with AHP.

6.2 Priorities among the M1 levers

Seven policy types are ranked, each corresponding to an M1 lever: capacity expansion (u_I), shore power (m_3), a daily cap (m_1), a higher fee (τ), rent subsidy through community investment (u_C), transit management (m_4), and alcohol control (m_2). The relevance $a_{vh} \in [0, 1]$ of lever v to feature h is shared by both destinations, so differences in the rankings come only from the pressure profiles. TOPSIS [11] weights each feature by its severity at destination d:

$$\omega_h^d = \frac{\omega_h (\tilde{F}_h^d)^2}{\sum_{h'} \omega_{h'} (\tilde{F}_{h'}^d)^2}, \qquad z_{vh}^d = \omega_h^d \frac{a_{vh}}{\sqrt{\sum_{v'} a_{v'h}^2}}, \qquad \mathrm{CC}_v^d = \frac{d_v^-}{d_v^+ + d_v^-}. \tag{M4-2}$$

Here d_v^+ and d_v^- are the distances from row z_v^d to the column-wise best and worst values. A closeness CC_v^d near 1 marks a lever that addresses the destination's dominant pressures; squaring $\tilde{F}_h^d$ concentrates attention on the most severe ones.



(a) Pressure profiles $\tilde{F}_h^d$ (b) Lever priorities CC_v^d

Figure 15: Juneau is nature-dominated and Venice urban- and social-dominated (a), so their lever priorities nearly reverse (b).

Juneau ranks shore power (0.758), capacity expansion (0.677), and transit management (0.512) highest. These relieve pressure without reducing volume, and match M1's choices of $m_3 = m_4 = 1$ with positive u_I. Alcohol control ranks last because TOPSIS ignores cost; M1 still adopts it because it costs only USD 0.1 million per year. Venice ranks the daily cap (0.626), rent subsidy (0.611), and higher fee (0.528) highest, which control demand and protect residents.
We perturbed every score by -1, 0, or $+1$ (clipped to 1–5) in 2,000 Monte Carlo draws and recomputed M4-1 and M4-2. Juneau's top two levers are unchanged in 95–96% of draws and Venice's in 62–64%. The order within Venice's cap–subsidy–fee cluster depends on the coding, but its first-ranked lever comes from that cluster in 94% of draws.

6.3 Recalibrating the M1 thresholds

Each M1 threshold θ is shifted by the feature gap between the destinations:

$$\theta^V = \mathrm{clip}\Big(\theta^J\Big[1 + \sum_h \xi_{\theta h}\,(\tilde{F}_h^V - \tilde{F}_h^J)\Big],\ \underline{\theta},\ \overline{\theta}\Big). \tag{M4-3}$$

The declared sensitivity $\xi_{\theta h}$ links a threshold only to the features it concerns. For example, $S_{\min}$ responds to F2, F3, and F7 with $\xi = 0.15, 0.20, 0.15$. The bounds keep each value within a plausible planning range.

M1 threshold	Symbol	Juneau θ^J	Factor	Venice θ^V
Acceptance target (M1-12)	$S_{\min}$	0.70	1.27	0.889
Social-pressure target (M1-12)	$H_{\max}$	1.00	0.78	0.780
Environmental ceiling (M1-9)	$E_{\max}$	1.00	0.83	0.830
Peak-day limit (M1-9), 10^4/day	x_{d0}	2.00	0.90	1.800
Revenue target, share of baseline (M1-12)	$R_{\min}/R_0$	0.92	0.90	0.828

Table 4: M1 thresholds recalibrated from Juneau to Venice.

Venice receives a stricter acceptance target and tighter crowding limits, and accepts a lower revenue target relative to its own baseline R_0^V. Its ecological ceiling also tightens, because the lagoon remains fragile even though wilderness pressure is low.

6.4 Evidence check on the tax channel

In M1 the tax lever τ lowers demand with elasticity $\varepsilon = -0.5$. Venice provides the only observed test of this channel. Its 5 EUR day-visitor access fee was charged on 29 days in 2024 and sold 485,062 vouchers for 2.43 million EUR. Daily sales averaged 16,726, with a peak of 24,686. [12] A 2024 survey gives the share $\pi(f)$ willing to pay a fee f: 0.680 at 5 EUR, falling to 0.065 at 50

EUR. [13] Anchoring this curve to the observed count gives vouchers $N(f)$ and gross revenue $G(f)$:

$$N(f) = N_5 \frac{\pi(f)}{\pi(5)}, \qquad G(f) = f\,N(f), \qquad N_5 = 485{,}062. \tag{M4-4}$$

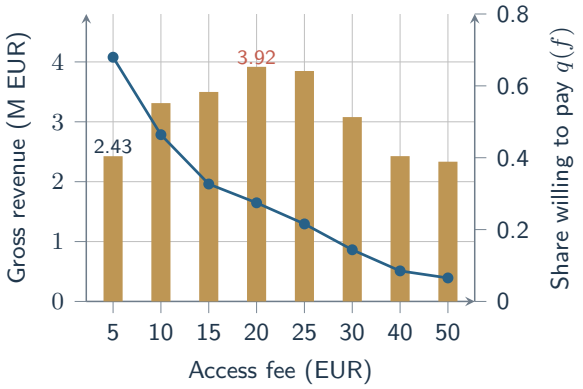


Figure 16: Bars: conditional gross fee revenue over the 29 charged days (left axis). Line: share willing to pay $\pi(f)$ (right axis).

Doubling the fee from 5 to 10 EUR cuts vouchers by 32%, an arc elasticity of -0.55, close to M1's assumed $\varepsilon = -0.5$. Gross revenue still rises 37%, so the fee acts as both demand control and funding, as in M1. Revenue peaks near 20 EUR at 3.92 million EUR (Figure 16).

This peak is conditional, not an optimum: the curve is hypothetical and from a small, self-selected sample. The only identified collection cost, a 229,871 EUR fixed service fee, caps net revenue at 2.20 million EUR at the 5 EUR anchor.

7 Model 5: Redistributing visitors among Juneau's attractions

M1 treats Juneau as a single node, but pressure concentrates at a few sites, especially the Mendenhall Glacier. M1 represents redistribution only through the switch m_5, which it leaves off: at the aggregate level, marketing simply adds demand ($\kappa_5 = +0.10$). M5 asks what redistribution achieves spatially. Can targeted investment in less-visited sites bring glacier utilization below 0.80 without reducing total visitors? It takes the M1 equilibrium $x = 147.2$ (10^4/year) and $x_d = 1.84$ (10^4/day) as given; 13 of the 18 preference vectors in M3 retain this volume.

7.1 Attraction nodes and gravity allocation

Visitors leave the port and choose among four nodes $i \in \mathcal{I}$ according to each node's attractiveness and its travel resistance from the port, as in a gravity (Huff) model [14]. The decay of demand with resistance reuses M1's elasticity ε.

Symbol	Meaning
$x_i,\ I_{cap,i}$	Annual visitors and annual capacity of node i, in 10^4 visitors; $\sum_i x_i = x$
$a_i,\ a_i^{\max}$	Effective and potential attractiveness of node i
$t_i,\ \varepsilon = -0.5$	Normalized travel resistance from the port; M1 demand elasticity

$$x_i = x\,\frac{a_i(1+t_i)^{\varepsilon}}{\sum_{i'} a_{i'}(1+t_{i'})^{\varepsilon}}. \tag{M5-1}$$

The allocation moves visitors between sites but never changes their total. Utilization $x_i/I_{cap,i}$ is the node-level counterpart of M1's capacity limit $x \le I_{cap}$. Resistances come from travel times of 0, 20, 35, and 40 minutes, and the baseline a_i reproduce the observed concentration at the port and glacier.

Node	Type	a_i / $a_i^{\max}$	t_i	$I_{cap,i}$	x_i	$x_i/I_{cap,i}$
N1 Port / downtown	hot	1.00 / 1.00	0.0	80.0	70.6	0.883
N2 Mendenhall Glacier	hot	0.79 / 0.79	0.4	49.6	47.1	0.950
N3 Whale watching	cold	0.33 / 0.75	0.7	35.0	17.9	0.511
N4 Rainforest	cold	0.22 / 0.45	0.8	28.0	11.6	0.414

Table 5: Node inputs and baseline allocation (x_i and $I_{cap,i}$ in 10^4 visitors/year). The glacier runs at 95% of capacity, while the cold sites remain below their potential appeal because they are less known and harder to reach.

7.2 Cold-site investment

USD 5 million per year goes to the cold sites: USD 3 million to whale watching and USD 2 million to the rainforest. As in M1, investment acts through response efficiencies η. Each node's amount is split equally between marketing u_i^a and transit u_i^t (10^4 USD):

$$\begin{aligned} a_i &= a_i^{\max} \min\Big(1,\ \tfrac{a_i^0}{a_i^{\max}} + \eta_a u_i^a\Big), \quad & t_i &= t_i^0 - \eta_t u_i^t, \\ I_{cap,i} &= I_{cap,i}^0 + \eta_I u_i^t, & b_i &= b_i^0 + \eta_b u_i^t. \end{aligned} \tag{M5-2}$$

Marketing raises awareness, moving a_i toward its potential. Transit spending shortens effective travel, expands site capacity, and raises the daily capacity b_i of the road or boat link from the port. The primary efficiencies $(\eta_a, \eta_t, \eta_I, \eta_b) = (0.003, 0.0013, 0.06, 0.0008)$ are assumptions, tested in Section 7.4.

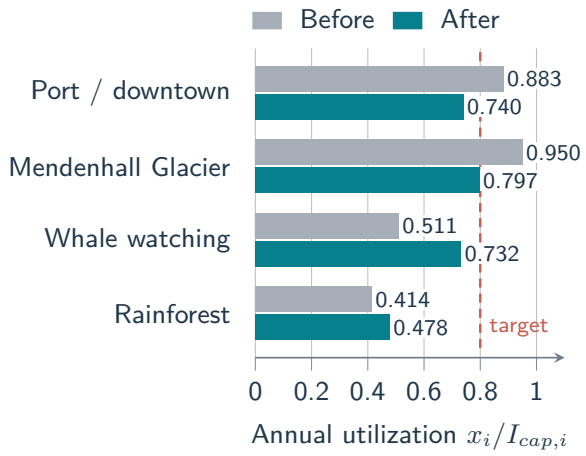


Figure 17: Node utilization before and after USD 5 million of cold-site investment.

Whale-watching attractiveness rises from 0.33 to 0.67 and rainforest attractiveness from 0.22 to 0.36. Their capacities grow to 44 and 34.

Glacier utilization falls from 0.950 to 0.797, below the 0.80 target, and the port falls from 0.883 to 0.740 (Figure 17). The cold sites absorb 19.0 (10^4) visitors: 11.4 from the port and 7.6 from the glacier. Both cold sites stay below 0.80, so the shift moves pressure without creating a new hotspot.

7.3 Peak-day network check

The reallocation must also work on M1's peak day. The peak x_d leaves the port over three links, one per site, with daily demand $x_{d,i} = x_d\, x_i/x$. The maximum flow [15] Q and its feasibility are

$$Q = \sum_{i=2}^{4} \min(b_i, x_{d,i}), \qquad \text{feasible if } Q = \sum_{i=2}^{4} x_{d,i}. \tag{M5-3}$$

	Before		After	
Link from port	$x_{d,i}$	b_i	$x_{d,i}$	b_i
To glacier (N2)	0.589	0.50	0.494	0.50
To whale watching (N3)	0.223	0.30	0.403	0.42
To rainforest (N4)	0.145	0.18	0.203	0.26
Maximum flow Q / total demand	0.868 / 0.957		1.100 / 1.100	

Table 6: Peak-day link demand and capacity (10^4 visitors/day).
Before investment, the port–glacier road is the binding cut: peak demand exceeds its capacity by 18%. Diversion relieves this road, and transit spending lets the cold-site links carry their larger flows. After investment, every link carries its full peak demand.

7.4 Sensitivity to response efficiencies

No data identify the efficiencies. The primary set was chosen so that USD 5 million reaches the 0.80 target, so 0.797 is a consistency check, not independent proof.

Efficiency set	$(\eta_a, \eta_t, \eta_I, \eta_b)$	**Glacier** $x_2/I_{cap,2}$	**Peak flow feasible**
Primary (target-calibrated)	(0.003, 0.0013, 0.06, 0.0008)	0.797	Yes
Neutral round values	(0.003, 0.0010, 0.06, 0.0010)	0.800	Yes
Conservative	(0.002, 0.0010, 0.05, 0.0005)	0.843	No
Optimistic	(0.004, 0.0020, 0.07, 0.0012)	0.757	Yes

Table 7: Glacier utilization under alternative response efficiencies.
The diversion and the port–glacier bottleneck appear under every set. The exact margin does not: neutral values place the glacier at the threshold, and conservative values leave it above. USD 5 million should therefore be treated as the expected scale, not a guaranteed amount. If observed glacier utilization remains above 0.80 after a season, transit funding for the cold-site links should be increased first.

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